PRACTICAL APPLICATION:

1. Threat Hunting Practice

Hypothesis Development:

In kali linux,

1) Check Ip of kali linux,

ip a

```
Session  Actions  Edit  View  Help

┌──(kali㉿kali)-[~]
└─$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group def
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP g
    link/ether 08:00:27:10:da:c8 brd ff:ff:ff:ff:ff:ff
    inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute eth0
       valid_lft 547sec preferred_lft 547sec
    inet6 fe80::a00:27ff:fe10:dac8/64 scope link noprefixroute
       valid_lft forever preferred_lft forever
```

Figure 1: ip address of kali linux

2) Then start Elastic search and access through browser,

sudo systemctl start elasticsearch
sudo systemctl start kibana
sudo systemctl start logstash
sudo systemctl start rsyslog

and check elastic search is working in kali linux

curl -k -u elastic https://10.0.2.15:9200

and access through browser using

https://10.0.2.15:5601

```
kali@kali: ~
Session  Actions  Edit  View  Help
└─$ sudo systemctl start elasticsearch

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start kibana

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start logstash

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start rsyslog

┌──(kali㉿kali)-[~]
└─$ sudo systemctl start ssh

┌──(kali㉿kali)-[~]
└─$ curl -k -u elastic https://10.0.2.15:9200
Enter host password for user 'elastic':
{
  "name" : "kali",
  "cluster_name" : "elasticsearch",
  "cluster_uuid" : "rW-EbySuQYCqsU82cuybNg",
  "version" : {
    "number" : "8.19.12",
    "build_flavor" : "default",
    "build_type" : "deb",
    "build_hash" : "840cd2a58b052d1632219ee0b8dcc0f364226287",
    "build_date" : "2026-02-23T23:08:40.713020893Z",
    "build_snapshot" : false,
    "lucene_version" : "9.12.2",
    "minimum_wire_compatibility_version" : "7.17.0",
    "minimum_index_compatibility_version" : "7.0.0"
  },
  "tagline" : "You Know, for Search"
}
```

Figure 2: started elastic search and it is working

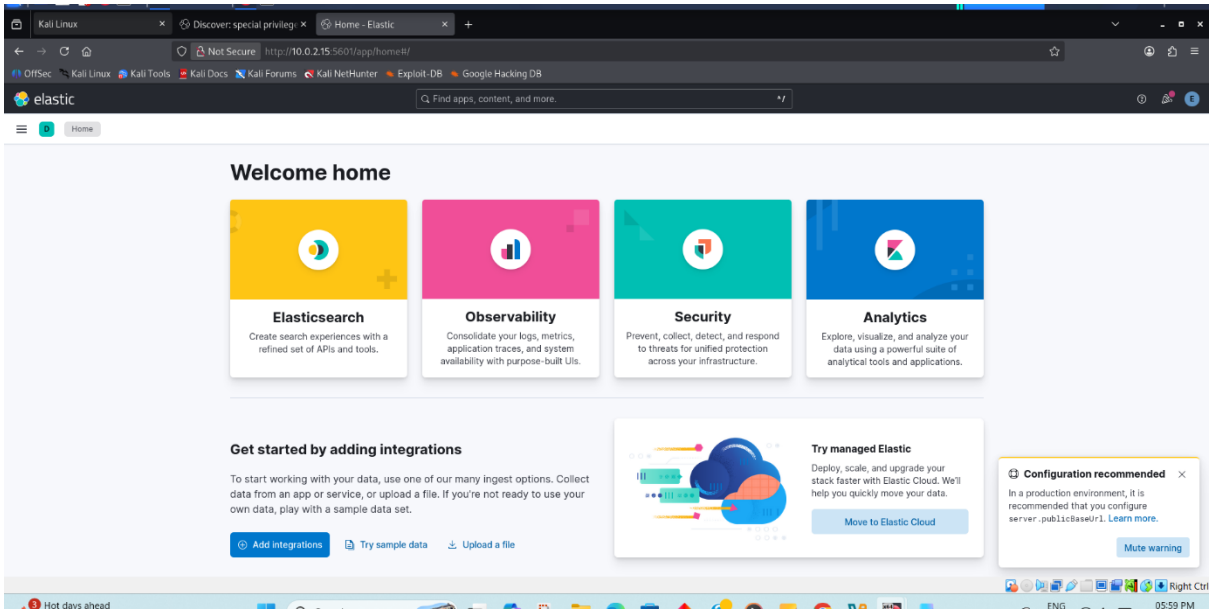


Figure 3: accessing through browser

Goto windows browser and check
http://10.0.2.15:5601

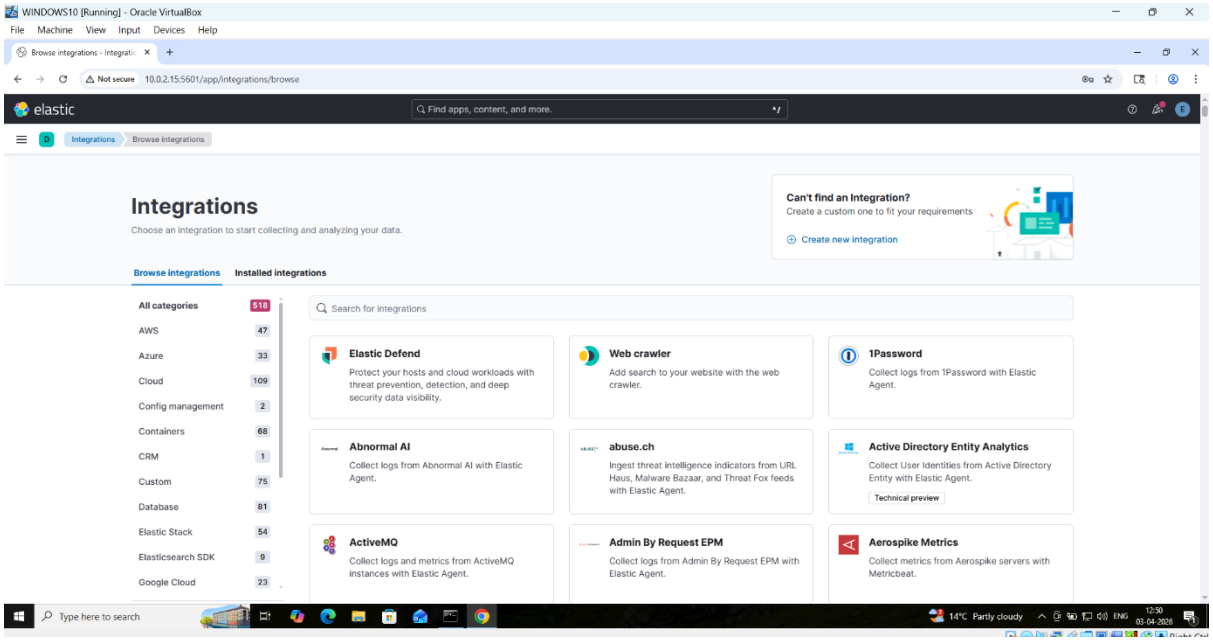


Figure 4: in windows elastic is accessible

3) Download and install Winlogbeat on Windows:
Download winlogbeat from official elastic download page

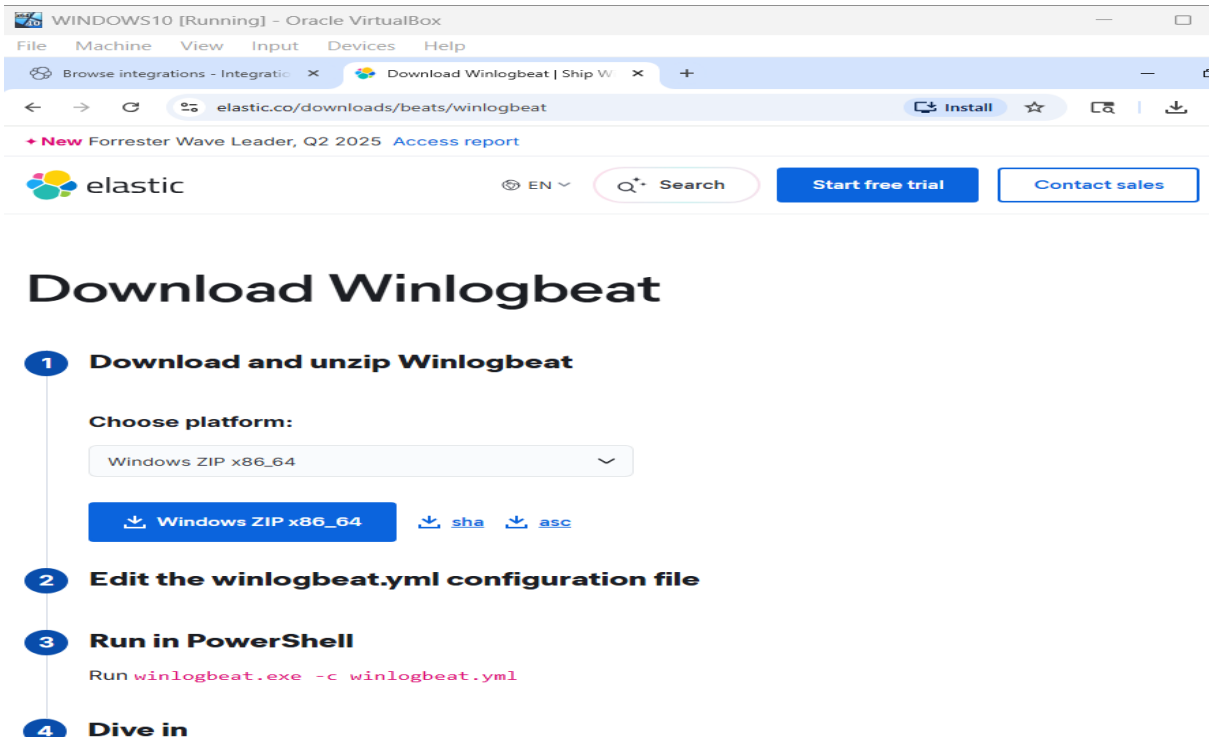


Figure 5: downloading winlogbeat on window

Then extract zip file and move it to
C:\Program Files\Winlogbeat

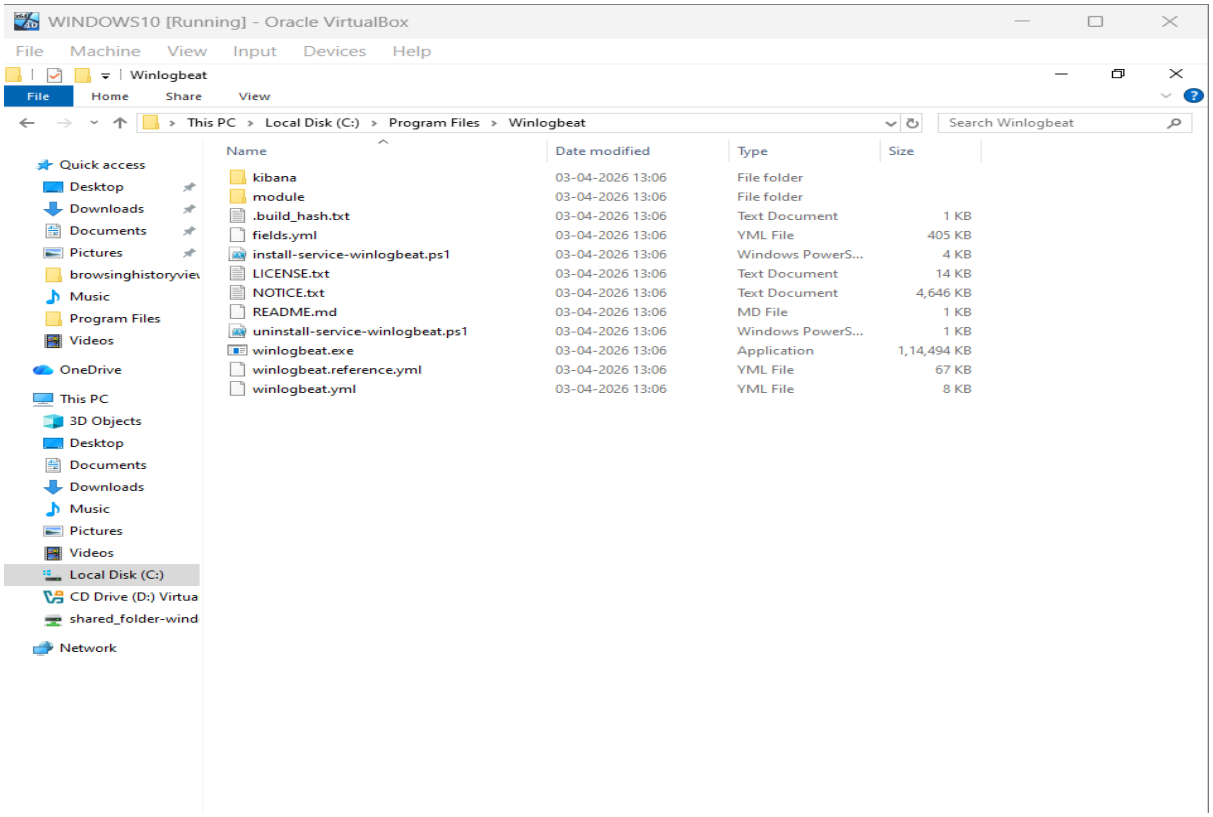


Figure 6: extracted winlogbeat file

4) Configure Winlogbeat:
Open Winlogbeat config file in Notepad as Administrator. Delete older content and add content as shown in below screenshot:

WINDOWS10 [Running] - Oracle VirtualBox

File Machine View Input Devices Help

*winlogbeat.yml - Notepad

File Edit Format View Help

```
winlogbeat.event_logs:
  - name: Security
  - name: System
  - name: Application

output.elasticsearch:
  hosts: ["https://10.0.2.15:9200"]
  username: "elastic"
  password: "9*3wyiB_sacrkOmV52nO"
  ssl.verification_mode: none

setup.kibana:
  host: "http://10.0.2.15:5601"
  username: "elastic"
  password: "9*3wyiB_sacrkOmV52nO"

processors:
  - add_host_metadata: ~
  - add_cloud_metadata: ~
```

Figure 7: configuring winlogbeat config file

5) Test the config before starting:
Open PowerShell as Administrator and goto

cd "C:\Program Files\Winlogbeat"

test the configuration is correct or not,

.\winlogbeat.exe test config -c .\winlogbeat.yml -e

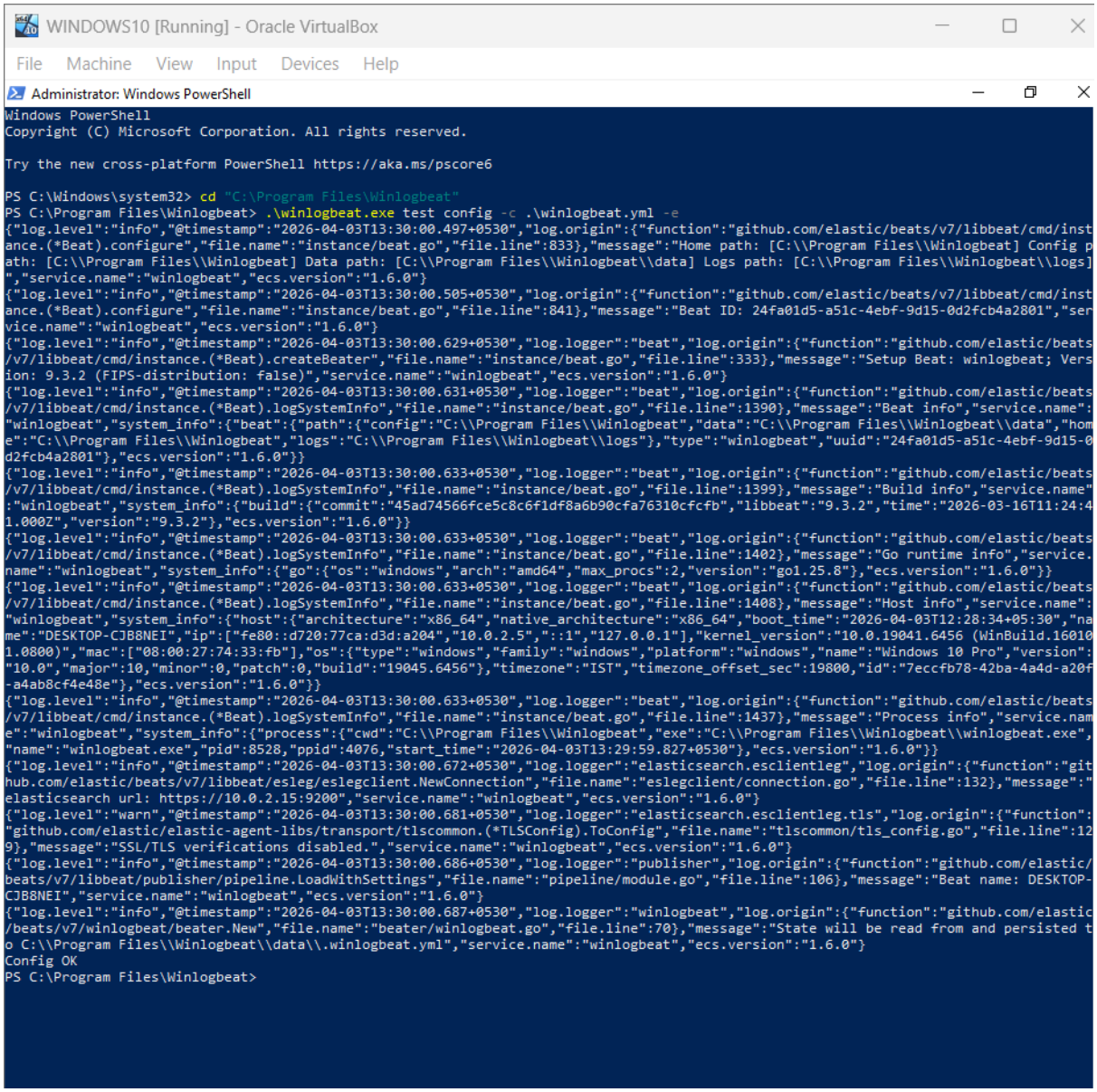


Figure 8: configuration is correct

6) Test connection to Elasticsearch:

Open PowerShell as Administrator and goto

```
cd "C:\Program Files\Winlogbeat"
```

test the config

```
.\winlogbeat.exe test config -c .\winlogbeat.yml -e
```

```
o C:\\Program Files\\Winlogbeat\\data\\.winlogbeat.yml","service.name":"winlogbeat","ecs.version":"1.6.0"}
Config OK
PS C:\Program Files\Winlogbeat> .\winlogbeat.exe test output -c .\winlogbeat.yml -e
{"log.level":"info","@timestamp":"2026-04-03T13:31:08.421+0530","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/ins
ance.(*Beat).configure","file.name":"instance/beat.go","file.line":833},"message":"Home path: [C:\\Program Files\\Winlogbeat] Config
ath: [C:\\Program Files\\Winlogbeat] Data path: [C:\\Program Files\\Winlogbeat\\data] Logs path: [C:\\Program Files\\Winlogbeat\\logs
","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-03T13:31:08.421+0530","log.origin":{"function":"github.com/elastic/beats/v7/libbeat/cmd/ins
ance.(*Beat).configure","file.name":"instance/beat.go","file.line":841},"message":"Beat ID: 24fa01d5-a51c-4ebf-9d15-0d2fcb4a2801","se
vice.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-03T13:31:08.446+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"gi
hub.com/elastic/beats/v7/libbeat/esleg/eslegclient.NewConnection","file.name":"eslegclient/connection.go","file.line":132},"message":
elasticsearch url: https://10.0.2.15:9200","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"warn","@timestamp":"2026-04-03T13:31:08.446+0530","log.logger":"elasticsearch.esclientleg.tls","log.origin":{"function"
"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":1
9},"message":"SSL/TLS verifications disabled.","service.name":"winlogbeat","ecs.version":"1.6.0"}
elasticsearch: https://10.0.2.15:9200...
  parse url... OK
  connection...
    parse host... OK
    dns lookup... OK
    addresses: 10.0.2.15
    dial up... OK
  TLS...{"log.level":"warn","@timestamp":"2026-04-03T13:31:08.486+0530","log.logger":"elasticsearch.esclientleg.tls","log.origin":{"f
nction":"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file
line":129},"message":"SSL/TLS verifications disabled.","service.name":"winlogbeat","ecs.version":"1.6.0"}

    security... WARN server's certificate chain verification is disabled
    handshake... OK
    TLS version: TLSv1.3
    dial up... OK
{"log.level":"warn","@timestamp":"2026-04-03T13:31:08.813+0530","log.logger":"elasticsearch.esclientleg.tls","log.origin":{"function"
"github.com/elastic/elastic-agent-libs/transport/tlscommon.(*TLSConfig).ToConfig","file.name":"tlscommon/tls_config.go","file.line":1
9},"message":"SSL/TLS verifications disabled.","service.name":"winlogbeat","ecs.version":"1.6.0"}
{"log.level":"info","@timestamp":"2026-04-03T13:31:08.921+0530","log.logger":"elasticsearch.esclientleg","log.origin":{"function":"gi
hub.com/elastic/beats/v7/libbeat/esleg/eslegclient.(*Connection).Ping","file.name":"eslegclient/connection.go","file.line":323},"mess
ge":"Attempting to connect to Elasticsearch version 8.19.12 (default)","service.name":"winlogbeat","ecs.version":"1.6.0"}
  talk to server... OK
  version: 8.19.12
PS C:\Program Files\Winlogbeat>
```

Figure 9: connection to elastic search is successful

7) Install Winlogbeat as Windows service:

In powershell ,

.\install-service-winlogbeat.ps1

Then start ,

Start-Service winlogbeat

Then check service,

Get-Service winlogbeat

```
PS C:\Program Files\Winlogbeat> Unblock-File -Path .\install-service-winlogbeat.ps1
PS C:\Program Files\Winlogbeat> .\install-service-winlogbeat.ps1

Status   Name               DisplayName
------   ----               -----------
Stopped  winlogbeat         winlogbeat


PS C:\Program Files\Winlogbeat> Start-Service winlogbeat
PS C:\Program Files\Winlogbeat> Get-Service winlogbeat

Status   Name               DisplayName
------   ----               -----------
Running  winlogbeat         winlogbeat


PS C:\Program Files\Winlogbeat>
```

Figure 10: installed winlogbeat as windows service

8) Goto elastic in kali linux and check if logs arrived .
Create a dataview in kibana, goto stack management → index pattern → click create index pattern and type

winlogbeat-*

choose time filter filed @timestamp
then save,

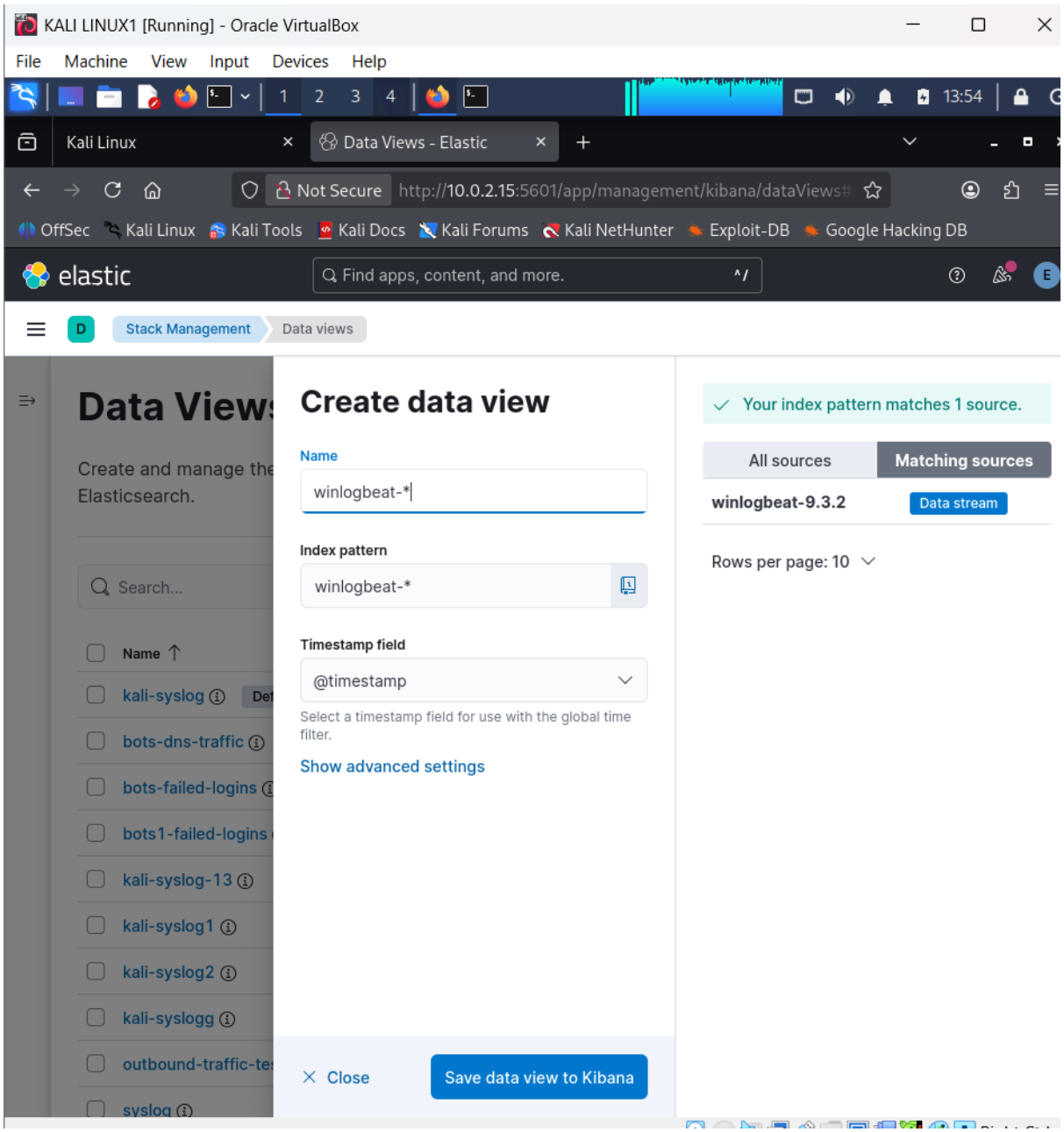


Figure 11: creating data view

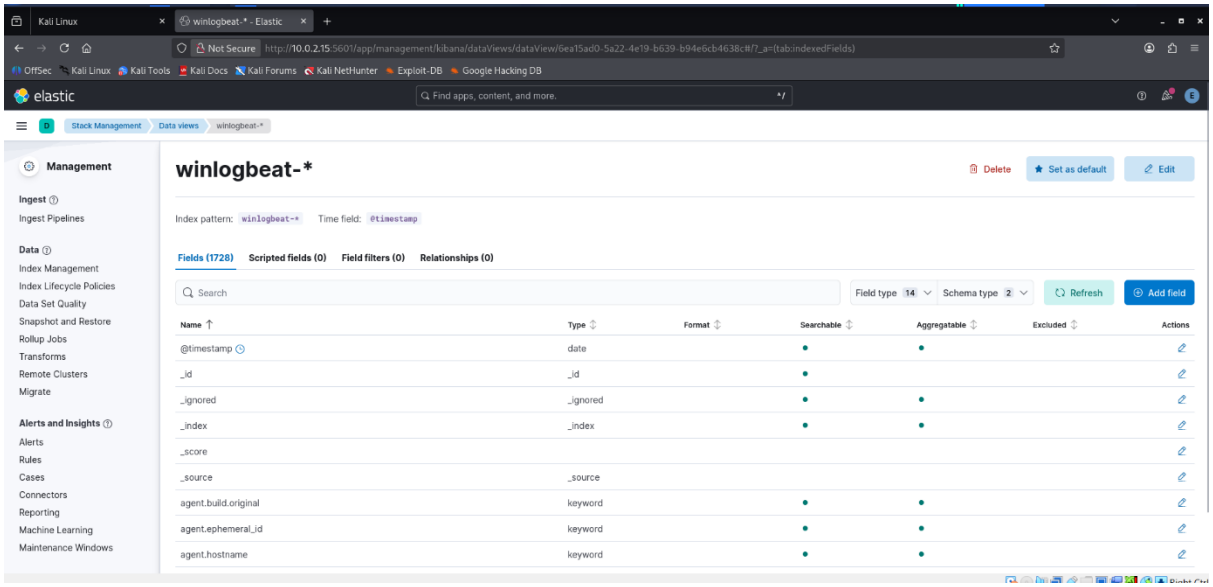


Figure 12: created dataview

9) then view in discover
Goto discover → then in dropdown select winlogbeat
To check failed login in search bar type,
Event.id:4625

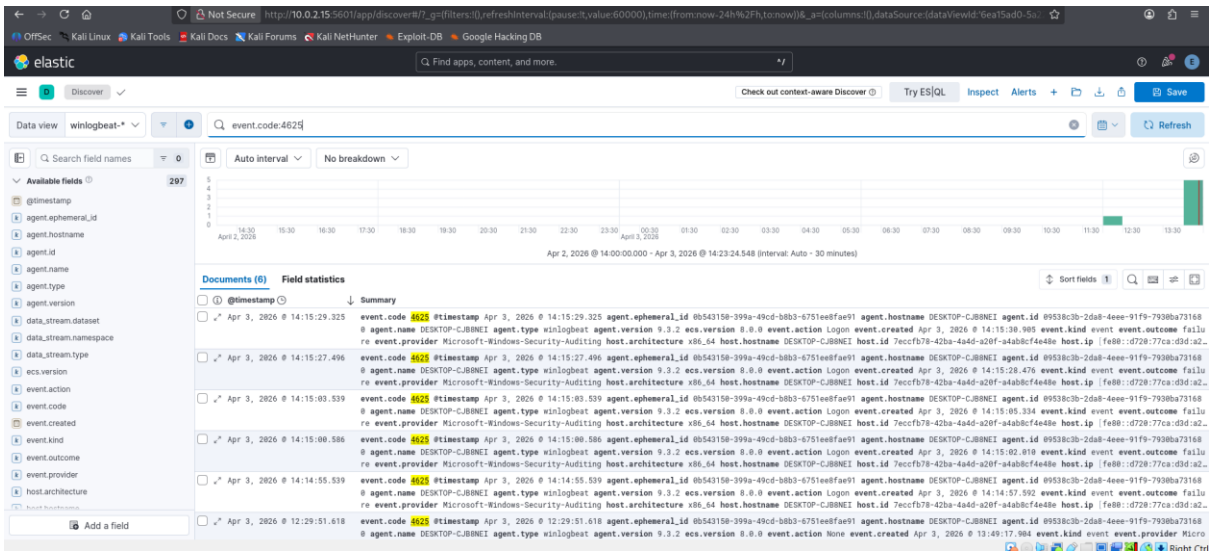


Figure 13: this shows failed login attempts , this confirms that in kibana logs from windows accessible

10) Formulate a hypothesis "Unauthorized privilege escalation in domain accounts" and Query Elastic Security for Event ID 4672 (role assignment) :

Hypothesis:

Unauthorized privilege escalation in domain accounts

Event ID 4672 indicates special privileges assigned to a user.

To find special privuleges assigned to user search in kibana event id 4672.
In kibana search bar type,

event.code:4672

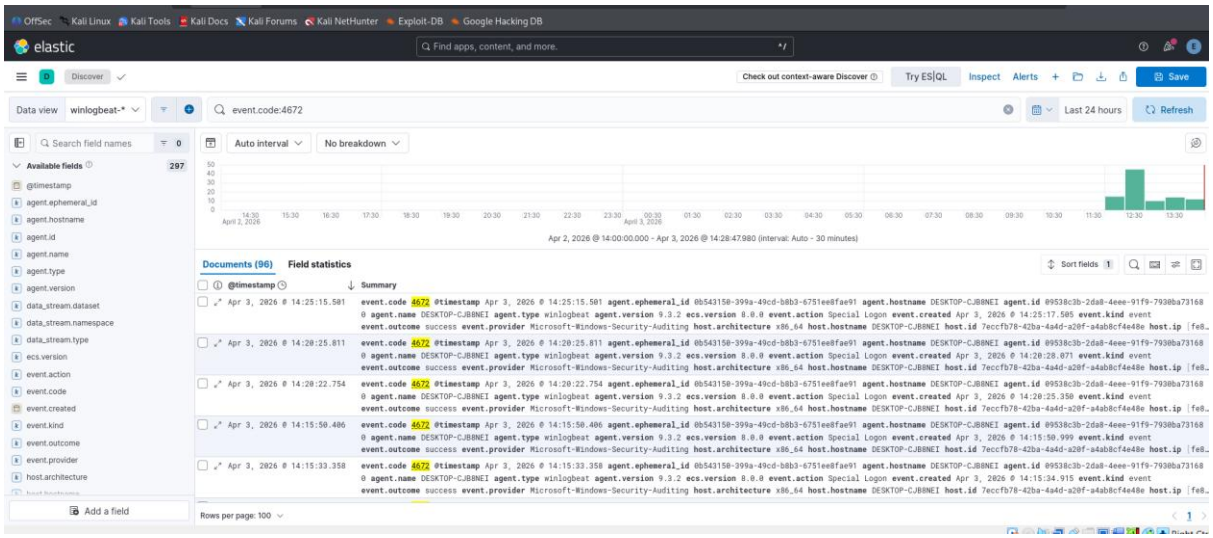


Figure 14: above the screenshot shows special privileges assigned to a user

Analyze the events,

Timestamp: Apr 3, 2026 @15:50:48
Username:SYSTEM
Hostname:DESKTOP-CJB8NEI
Event code:4672
Message:special privileges assigned to new logon
Host od type: windows

Multiple Event ID 4672 logs were observed in Elastic Security.
These events indicate that special privileges were assigned to newly logged-on accounts.

Document:

Timestamp	user	Event ID	Notes
Apr 3, 2026 @ 15:50:48	System	4672	Special privileges assigned to new logon

Multiple repeated Event ID 4672 events were identified on the Windows host DESKTOP-CJB8NEI. These events indicate that special privileges were assigned to newly logged-on accounts.

Threat Intelligence Hunt:

Use AlienVault OTX to search for T1078 IOCs:
In alienvault OTX searched for T1078 IOcs and collected meta data
Pulse name: oz batch 50 IOCs
Created 2 months ago
TLP white
IOC count 50
Average BDE score 80

Collected example data:

IOC type	indicator	source
Ipv4	205.210.31.161	alienvault otx
Ipv4	205.210.31.162	alienvaultotx
Ipv4	205.210.31.163	alienvaultotx

AlienVault OTX was used to search for indicators of compromise related to MITRE ATT&CK T1078 – Valid Accounts.
These IP addresses may be suspicious and can be used as threat intelligence indicators.
They may be related to unauthorized access or suspicious remote activity.
collected them to analyze Event ID 4672 and possible MITRE ATT&CK T1078 Valid Accounts activity.

Cross-reference with Velociraptor queries:
For SELECT * FROM processes it showed Velociraptor does not have a plugin named processes in current query scope so used query is

```
SELECT Name, Pid, Exe, CommandLine, Username
FROM pslist()
```

This also shows process name, PID, executable path, command line and user.

In velociraptor goto notebook → click on notebook → give name → then click on launch
After that add query cell → select VQL and select edit cell and paste

```
SELECT Name, Pid, Exe, CommandLine, Username
FROM pslist()
```

After that click save and run

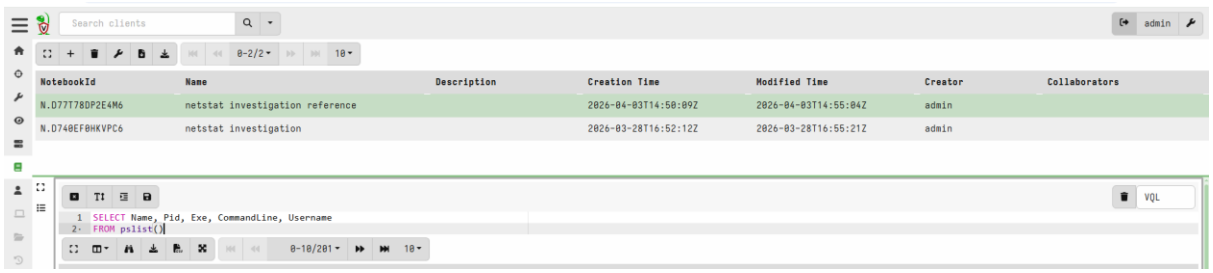


Figure 15: running query

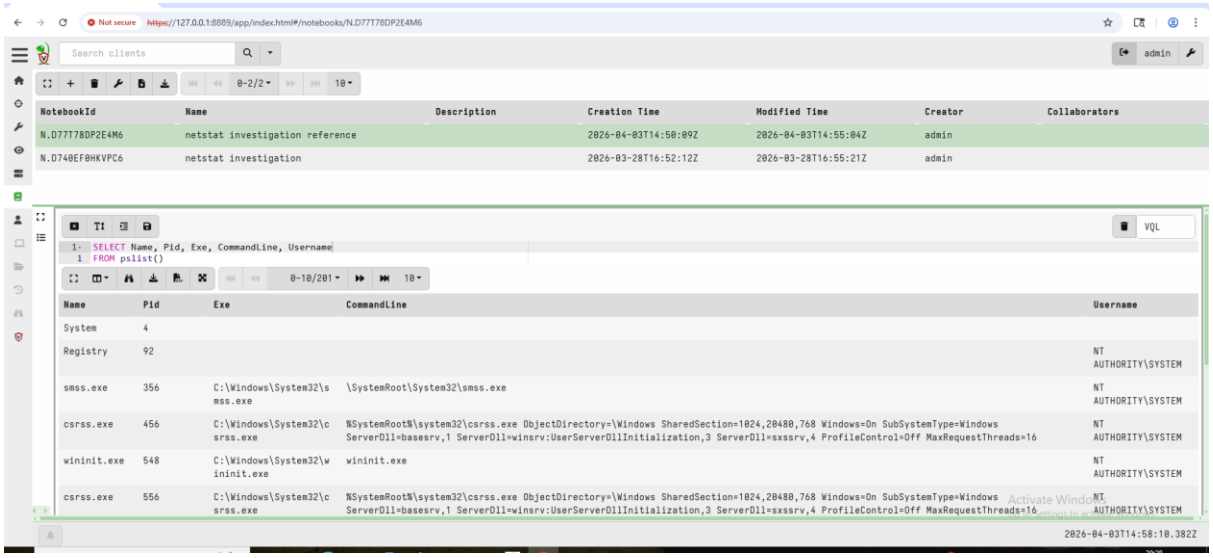
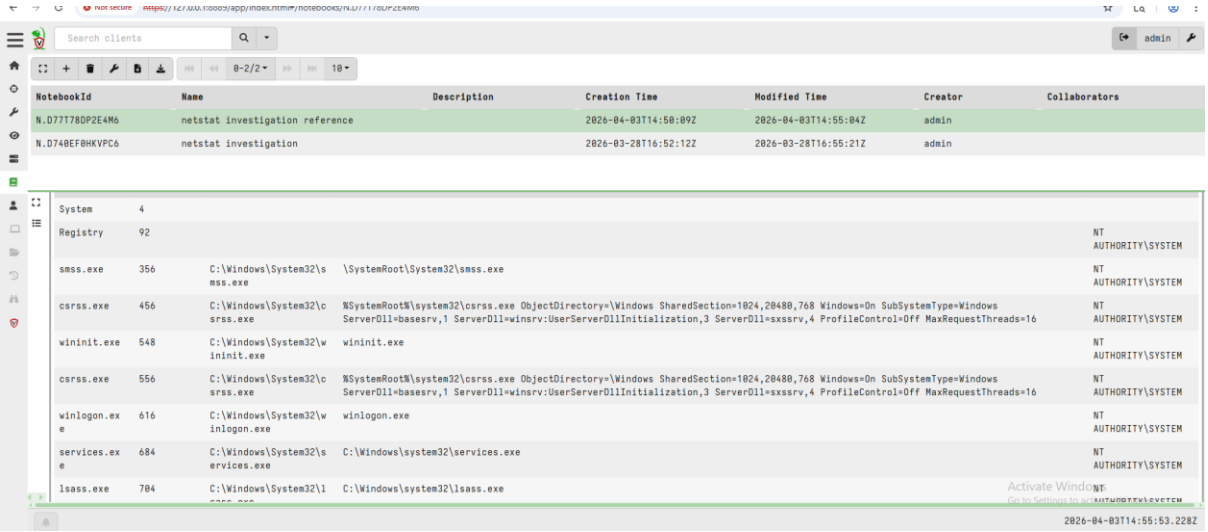


Figure 16

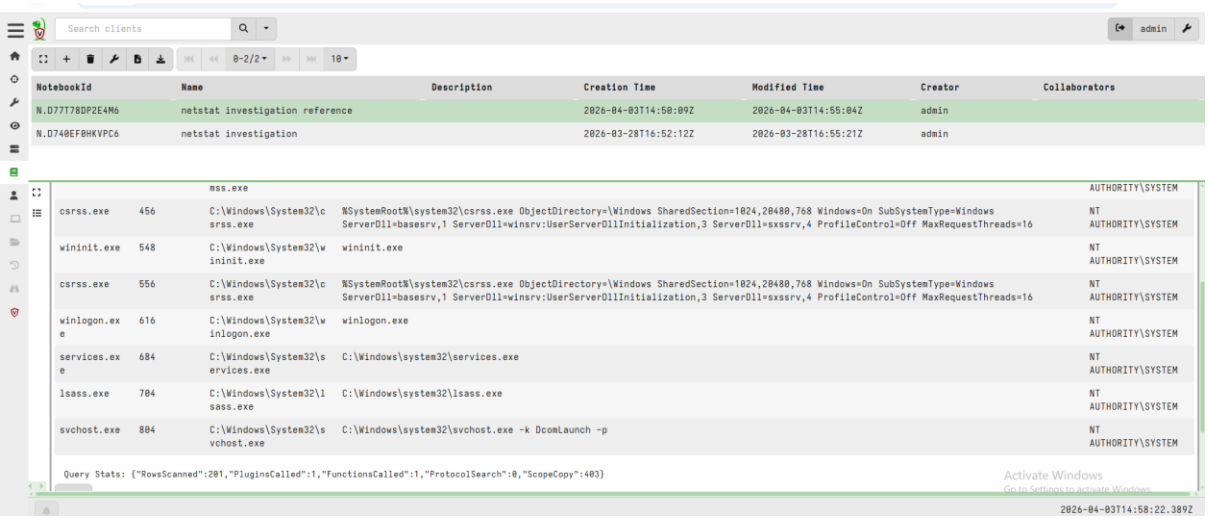


NotebookId	Name	Description	Creation Time	Modified Time	Creator	Collaborators
N.D77T78DP2E4M6	netstat investigation reference		2026-04-03T14:50:09Z	2026-04-03T14:55:04Z	admin	
N.D740EF0HKVPC6	netstat investigation		2026-03-28T16:52:12Z	2026-03-28T16:55:21Z	admin	

System	4			
Registry	92			NT AUTHORITY\SYSTEM
smss.exe	356	C:\Windows\System32\smss.exe	\SystemRoot\System32\smss.exe	NT AUTHORITY\SYSTEM
csrss.exe	456	C:\Windows\System32\csrss.exe	%SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16	NT AUTHORITY\SYSTEM
wininit.exe	548	C:\Windows\System32\wininit.exe	wininit.exe	NT AUTHORITY\SYSTEM
csrss.exe	556	C:\Windows\System32\csrss.exe	%SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16	NT AUTHORITY\SYSTEM
winlogon.exe	616	C:\Windows\System32\winlogon.exe	winlogon.exe	NT AUTHORITY\SYSTEM
services.exe	684	C:\Windows\System32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM
lsass.exe	704	C:\Windows\System32\lsass.exe	C:\Windows\system32\lsass.exe	NT

2026-04-03T14:55:53.228Z

Figure 17



NotebookId	Name	Description	Creation Time	Modified Time	Creator	Collaborators
N.D77T78DP2E4M6	netstat investigation reference		2026-04-03T14:50:09Z	2026-04-03T14:55:04Z	admin	
N.D740EF0HKVPC6	netstat investigation		2026-03-28T16:52:12Z	2026-03-28T16:55:21Z	admin	

		mss.exe		AUTHORITY\SYSTEM
csrss.exe	456	C:\Windows\System32\csrss.exe	%SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16	NT AUTHORITY\SYSTEM
wininit.exe	548	C:\Windows\System32\wininit.exe	wininit.exe	NT AUTHORITY\SYSTEM
csrss.exe	556	C:\Windows\System32\csrss.exe	%SystemRoot%\system32\csrss.exe ObjectDirectory=\Windows SharedSection=1024,20480,768 Windows=On SubSystemType=Windows ServerDll=basesrv,1 ServerDll=winsrv:UserServerDllInitialization,3 ServerDll=sxssrv,4 ProfileControl=Off MaxRequestThreads=16	NT AUTHORITY\SYSTEM
winlogon.exe	616	C:\Windows\System32\winlogon.exe	winlogon.exe	NT AUTHORITY\SYSTEM
services.exe	684	C:\Windows\System32\services.exe	C:\Windows\system32\services.exe	NT AUTHORITY\SYSTEM
lsass.exe	704	C:\Windows\System32\lsass.exe	C:\Windows\system32\lsass.exe	NT AUTHORITY\SYSTEM
svchost.exe	804	C:\Windows\System32\svchost.exe	C:\Windows\system32\svchost.exe -k DcomLaunch -p	NT AUTHORITY\SYSTEM

Query Stats: {"RowsScanned":201,"PluginsCalled":1,"FunctionsCalled":1,"ProtocolSearch":0,"ScopeCopy":403}

2026-04-03T14:58:22.389Z

Figure 18

Figure 17,18,19 The observed processes appear to be normal Windows operating system processes.
No clearly suspicious or attacker-named process was identified in the visible output.

The threat hunt was conducted by combining AlienVault OTX threat intelligence with Velociraptor endpoint process analysis. The OTX search provided suspicious IP indicators related to possible T1078 Valid Accounts activity, while Velociraptor was used to review running processes on the endpoint. The endpoint showed mostly normal Windows processes, and no direct malicious process was identified.

Hunting Report:

Summarizing findings and mapping to MITRE ATT&CK T1078:

Elastic Security was used to review Event ID 4672, which indicates that special privileges were assigned during a logon session. Multiple events were observed, mainly involving the SYSTEM account. Since attackers may abuse valid accounts to gain privileged access, this event was selected for further analysis.

To support the hunt, AlienVault OTX was used to search for suspicious indicators related to T1078. Example IP addresses were collected from a public pulse and used as threat intelligence.

These indicators were then cross-referenced with endpoint activity using Velociraptor. A process query was run to inspect active processes on the Windows system. The results showed mostly normal Windows processes. No direct malicious process was identified .

These findings are mapped to MITRE ATT&CK T1078 – Valid Accounts, because the activity involves the use of legitimate accounts and privileged logon behaviour that could be abused by an attacker.